



FACULTY OF COMPUTER SCIENCE AND MATHEMATICS

CSF3233: (K2)

SEMESTER 2 SESSION 2025/2026

LAB REPORT 5

LAB 5: NETWORK SECURITY

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**PROGRAM NAME: BACHELOR OF COMPUTER SCIENCE (SOFTWARE
ENGINEERING) (HONS)**

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TASK 1: SETTING UP THE VIRTUAL LAB NETWORK

OBJECTIVE

To set up the network based on the provided topology.

TASK DESCRIPTION

In this task, the student needs to set up the network according to the given topology. This network will be used in the next task.

ESTIMATED TIME

30 Minutes

STEPS:

Output:

```
(kali@kali)-[~]
$ ping 192.168.1.4
PING 192.168.1.4 (192.168.1.4) 56(84) bytes of data:
64 bytes from 192.168.1.4: icmp_seq=1 ttl=64 time=2.76 ms
64 bytes from 192.168.1.4: icmp_seq=2 ttl=64 time=0.712 ms
64 bytes from 192.168.1.4: icmp_seq=3 ttl=64 time=0.842 ms
64 bytes from 192.168.1.4: icmp_seq=4 ttl=64 time=0.661 ms
64 bytes from 192.168.1.4: icmp_seq=5 ttl=64 time=1.53 ms
^C
--- 192.168.1.4 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4023ms
rtt min/avg/max/mdev = 0.661/1.299/2.755/0.791 ms
(kali@kali)-[~]
$
```

```
C:\Documents and Settings\user>ping 192.168.1.4
Pinging 192.168.1.4 with 32 bytes of data:
Reply from 192.168.1.4: bytes=32 time=2ms TTL=64
Reply from 192.168.1.4: bytes=32 time=1ms TTL=64
Reply from 192.168.1.4: bytes=32 time=1ms TTL=64
Reply from 192.168.1.4: bytes=32 time<1ms TTL=64

Ping statistics for 192.168.1.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 2ms, Average = 1ms
```

TASK 2: MAN-IN-THE-MIDDLE ATTACK USING ARP POISONING

OBJECTIVE

To run a man-in-the-middle (MITM) attack simulation using the ARP poisoning technique.

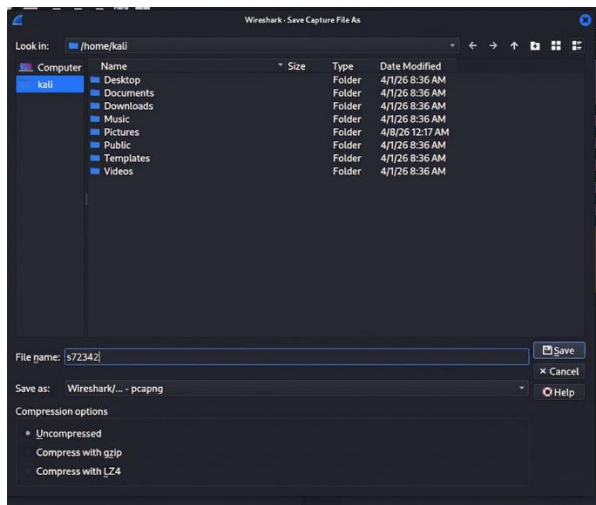
TASK DESCRIPTION

For this task, the student will run a man-in-the-middle attack simulation. This attack will use the ARP poisoning technique where all the network packets from the victim will be captured by using a sniffer software known as Wireshark. Before that, the ARP poisoning/spoofing will be carried out using a special tool in Kali Linux. This tool is known as Ettercap. By using Ettercap, all the packets during the telnet connection between Windows XP (victim) and Metasploitable will be redirected to Kali Linux (attacker) machine as well.

ESTIMATED TIME

90 Minutes

STEPS:



REFLECTION QUESTIONS

1. In your own words, explain ARP poisoning.
2. How can we prevent the ARP poisoning attack?
3. Why do you think telnet is not a safe way to connect to a remote machine? What is the safer way to do so?

1. In your own words, explain ARP poisoning.

Basically, ARP poisoning is when someone on a local network sends out fake messages to trick devices into thinking the attacker's computer is the router or another specific server. Since the protocol (ARP) is pretty trusting, the victim's computer updates its "address book" with the attacker's info. This lets the attacker sit in the middle and see all the data passing through before it even gets to the real destination.

2. How can we prevent the ARP poisoning attack?

The best way is to use Dynamic ARP Inspection (DAI) on network switches, which basically checks every ARP message to make sure it's legit before letting it through. You could also use Static ARP entries for super important devices so their addresses can't be changed by fake messages. Also, just using a VPN or HTTPS helps because even if the ARP is poisoned, the data itself is still encrypted and unreadable.

3. Why do you think telnet is not a safe way to connect to a remote machine? What is the safer way to do so?

Telnet is dangerous because it doesn't use any encryption at all. Everything you type including your login and password is sent in plain text. If an attacker is sniffing the network, they can see exactly what you're doing. The much safer way is to use SSH (Secure Shell). It does the same thing as Telnet but encrypts the whole connection, so your credentials and data stay private.

TASK 3: ANALYSING THE NETWORK PACKET

OBJECTIVE

To analyse the network packet using multiple tools

TASK DESCRIPTION

For this task, student needs to run packet analysis tools from various sample taken from the provided website.

ESTIMATED TIME

60 Minutes

STEPS:

```
kali@kali: ~/S72342
Session Actions Edit View Help
zsh: corrupt history file /home/kali/.zsh_history
(kali@kali)-[~]
$ cd S72342
(kali@kali)-[~/S72342]
$ conference.pcapng telnet-cooked.pcap
conference.pcapng: command not found
(kali@kali)-[~/S72342]
$ ls
conference.pcapng  telnet-cooked.pcap
(kali@kali)-[~/S72342]
$ dsniff -p telnet-cooked.pcap
dsniff: magic: unknown decode: 'pop'
dsniff: magic: unknown decode: 'pop'
dsniff: using telnet-cooked.pcap
>>>> 04/28/26 23:32:18 tcp 192.168.0.2:1550 -> 192.168.0.1:23 (telnet)
fake
user
/sbin/ping www.yahoo.com
ls
ls -a
exit
(kali@kali)-[~/S72342]
$
```

```
Session Actions Edit View Help
$ tshark -r conference.pcapng -Y http.request -T fields -e http.host -e htt
p.user_agent |sort -u|uniq -c|sort -n
  1 9gag.com Mozilla/5.0 (Windows NT 6.1; WOW64) AppleWebKit/537.3
6 (KHTML, like Gecko) Chrome/41.0.2272.89 Safari/537.36
  1 adx.g.doubleclick.net Mozilla/5.0 (Windows NT 6.1; WOW64) AppleWebK
it/537.36 (KHTML, like Gecko) Chrome/41.0.2272.89 Safari/537.36
  1 ajax-9gag-lol.9cache.com Mozilla/5.0 (Windows NT 6.1; WOW64) A
ppleWebKit/537.36 (KHTML, like Gecko) Chrome/41.0.2272.89 Safari/537.36
  1 ajax.googleapis.com Mozilla/5.0 (Windows NT 6.1; WOW64) AppleWebK
it/537.36 (KHTML, like Gecko) Chrome/41.0.2272.89 Safari/537.36
  1 ams1.ib.adnxs.com Mozilla/5.0 (Windows NT 6.1; WOW64) AppleWebK
it/537.36 (KHTML, like Gecko) Chrome/41.0.2272.89 Safari/537.36
  1 assets-9gag-ftw.9cache.com Mozilla/5.0 (Windows NT 6.1; WOW64) A
ppleWebKit/537.36 (KHTML, like Gecko) Chrome/41.0.2272.89 Safari/537.36
  1 a.thumbs.redditmedia.com Mozilla/5.0 (Windows NT 6.1; WOW64) A
ppleWebKit/537.36 (KHTML, like Gecko) Chrome/41.0.2272.89 Safari/537.36
  1 b.thumbs.redditmedia.com Mozilla/5.0 (Windows NT 6.1; WOW64) A
ppleWebKit/537.36 (KHTML, like Gecko) Chrome/41.0.2272.89 Safari/537.36
  1 cdn.adnxs.com Mozilla/5.0 (Windows NT 6.1; WOW64) AppleWebKit/537.3
6 (KHTML, like Gecko) Chrome/41.0.2272.89 Safari/537.36
  1 cdn.tradelab.fr Mozilla/5.0 (Windows NT 6.1; WOW64) AppleWebKit/537.3
6 (KHTML, like Gecko) Chrome/41.0.2272.89 Safari/537.36
  1 connect.facebook.net Mozilla/5.0 (Windows NT 6.1; WOW64) AppleWebK
it/537.36 (KHTML, like Gecko) Chrome/41.0.2272.89 Safari/537.36
  1 csi.gstatic.com Mozilla/5.0 (Windows NT 6.1; WOW64) AppleWebKit/537.3
6 (KHTML, like Gecko) Chrome/41.0.2272.89 Safari/537.36
  1 c.thumbs.redditmedia.com Mozilla/5.0 (Windows NT 6.1; WOW64) A
ppleWebKit/537.36 (KHTML, like Gecko) Chrome/41.0.2272.89 Safari/537.36
```

```

Session Actions Edit View Help
└─$ tshark -r conference.pcapng --export-objects http,tshark_folder
  1  0.000000 172.16.254.128 → 216.58.208.206 SSL 55
  2  0.000363 216.58.208.206 → 172.16.254.128 TCP 60 443 → 52166 [ACK] Seq=
=1 Ack=2 Win=64240 Len=0
  3  0.289945 172.16.254.128 → 216.58.208.226 TCP 55 51667 → 80 [ACK] Seq=
1 Ack=1 Win=63142 Len=1
  4  0.293835 216.58.208.226 → 172.16.254.128 TCP 60 80 → 51667 [ACK] Seq=
1 Ack=2 Win=64240 Len=0
  5  3.464487 172.16.254.128 → 8.8.8.8      DNS 73 Standard query 0xae84 A
www.google.fr
  6  3.482495      8.8.8.8 → 172.16.254.128 DNS 89 Standard query response
0xae84 A www.google.fr A 216.58.208.227
  7  3.483014 172.16.254.128 → 216.58.208.227 TCP 66 52182 → 443 [SYN] Seq
=0 Win=8192 Len=0 MSS=1460 WS=256 SACK_PERM
  8  3.483412 172.16.254.128 → 216.58.208.227 TCP 66 52183 → 443 [SYN] Seq
=0 Win=8192 Len=0 MSS=1460 WS=256 SACK_PERM
  9  3.495475 216.58.208.227 → 172.16.254.128 TCP 60 443 → 52182 [SYN, ACK
] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460
 10  3.495533 172.16.254.128 → 216.58.208.227 TCP 54 52182 → 443 [ACK] Seq
=1 Ack=1 Win=64240 Len=0
 11  3.496481 172.16.254.128 → 216.58.208.227 TLSv1 571 Client Hello (SNI=
www.google.fr)
 12  3.496735 216.58.208.227 → 172.16.254.128 TCP 60 443 → 52182 [ACK] Seq
=1 Ack=518 Win=64240 Len=0
 13  3.499742 VMware_b6:b5:48 → Broadcast   ARP 42 Who has 172.16.21.128?
Tell 172.16.254.128
 14  3.502024 216.58.208.227 → 172.16.254.128 TCP 60 443 → 52183 [SYN, ACK
] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460

```

```

kali@kali: ~/S72342
Session Actions Edit View Help
└─(kali@kali)~[~/S72342]
└─$ ls -l tshark_folder
total 10644
-rw-r--r-- 1 kali kali    32098 Apr 28 23:33 13123798753128404613
-rw-r--r-- 1 kali kali    28368 Apr 28 23:33 13734329343262018815
-rw-r--r-- 1 kali kali    17944 Apr 28 23:33 1426571640.037_Yqewu3_300.jpg
-rw-r--r-- 1 kali kali    26134 Apr 28 23:33 1426572438.5952_AzemY5_300.jpg
-rw-r--r-- 1 kali kali    12802 Apr 28 23:33 1426572509.5288_GA3EHE_300.jpg
-rw-r--r-- 1 kali kali    13751 Apr 28 23:33 1426572551.239_HA4Ury_300.jpg
-rw-r--r-- 1 kali kali    16211 Apr 28 23:33 1426572606.7525_YHYsYh_300.jpg
-rw-r--r-- 1 kali kali    12738 Apr 28 23:33 1426572618.4293_uMYhez_300.jpg
-rw-r--r-- 1 kali kali    23508 Apr 28 23:33 1426572637.302_XeGaPE_300.jpg
-rw-r--r-- 1 kali kali    17342 Apr 28 23:33 1426572669.1706_hYgeMA_300.jpg
-rw-r--r-- 1 kali kali    15318 Apr 28 23:33 1426572694.0224_UlyrA6_300.jpg
-rw-r--r-- 1 kali kali    12773 Apr 28 23:33 1426608057.9397_vYmeTE_300.jpg
-rw-r--r-- 1 kali kali    114685 Apr 28 23:33 %2f
-rw-r--r-- 1 kali kali    79635 Apr 28 23:33 '%2f(1)\'
-rw-r--r-- 1 kali kali    6469 Apr 28 23:33 '%2f(2)\'
-rw-r--r-- 1 kali kali     43 Apr 28 23:33 '%3ftype=imp5cnt=26location=http%
3A%2F%2Fgag.com%2F5ua=Mozilla%2F5.0%20(Windows%20NT%206.1%3B%20WOW64)%20Appl
eWebKit%2F537.36%20(KHTML%2CK%20like%20Gecko)%20.1545zip=750116afq=06rmk=06seg
=6city=Paris5state=FR%3AA88seller=1816site=554186pub=179096sz=300*250'
-rw-r--r-- 1 kali kali    36633 Apr 28 23:33 533a1088a4c24493adeef8152b98913a
.jpg
-rw-r--r-- 1 kali kali    28947 Apr 28 23:33 696ee4d325834ea0ab15220bfd7899cf
.jpg
-rw-r--r-- 1 kali kali    29611 Apr 28 23:33 '6Dg4oLk8bYq.js%3fversion=41'

```

```

kali@kali: ~/S72342
Session Actions Edit View Help
-rw-r--r-- 1 kali kali    117182 Apr 28 23:39 'widgets(1).js'
-rw-r--r-- 1 kali kali    117182 Apr 28 23:39 widgets.js
-rw-r--r-- 1 kali kali     2509 Apr 28 23:39 WnWL9EiaP8UCXn12wtaLaNr05QgM4Dfr
tammyb88RMU(1).png
-rw-r--r-- 1 kali kali     2509 Apr 28 23:39 WnWL9EiaP8UCXn12wtaLaNr05QgM4Dfr
tammyb88RMU.png
-rw-r--r-- 1 kali kali     351 Apr 28 23:39 'xaM9crmp-ArFY0RT(1).png'
-rw-r--r-- 1 kali kali     351 Apr 28 23:39 xaM9crmp-ArFY0RT.png
-rw-r--r-- 1 kali kali     145 Apr 28 23:39 'x_button_blue2(1).png'
-rw-r--r-- 1 kali kali     145 Apr 28 23:39 x_button_blue2.png
-rw-r--r-- 1 kali kali     4064 Apr 28 23:39 'YaWJC69hN-ZUnzy2(1).jpg'
-rw-r--r-- 1 kali kali     4064 Apr 28 23:39 YaWJC69hN-ZUnzy2.jpg
-rw-r--r-- 1 kali kali     6179 Apr 28 23:39 zg1lzZecwrY5ehR7_WAcg0zNcvq4lNHq
ld2YTx3FbJE(1).jpg
-rw-r--r-- 1 kali kali     6179 Apr 28 23:39 zg1lzZecwrY5ehR7_WAcg0zNcvq4lNHq
ld2YTx3FbJE.jpg
└─(kali@kali)~[~/S72342]
└─$ urlsnarf -p conference.pcapng | grep *http://* | cut -d "/" -f 5
urlsnarf: using conference.pcapng [tcp port 80 or port 8080 or port 3128]
└─(kali@kali)~[~/S72342]
└─$ urlsnarf -p conference.pcapng | grep *http://* | cut -d "/" -f 6 | sort -u
urlsnarf: using conference.pcapng [tcp port 80 or port 8080 or port 3128]
└─(kali@kali)~[~/S72342]
└─$

```


REFLECTION QUESTIONS

1. Based on your understanding, what is a network security and what are the basic objectives of network security?
2. Differentiate between Intrusion Detection System (IDS) and Intrusion Prevention System (IPS).
3. Why Virtual Private Network (VPN) is important for network security?
4. In what situation you might want to install a Demilitarized Zone (DMZ) on your network?
5. What is the role of Network Address Translation (NAT) on a network?
6. To applied and manage the security principle, you must understand how the network devices and technologies operate. Explain how the proxy server operates to provide security on the network.

1. What is network security and its basic objectives?

Network security refers to the policies, processes, and technologies designed to protect a network and its data from unauthorized access, misuse, or theft.

The basic objectives are known as the CIA Triad:

- Confidentiality: Ensuring that sensitive information is accessible only to those authorized to have access.
- Integrity: Maintaining the consistency and accuracy of data by ensuring it isn't modified during transit.
- Availability: Ensuring that the network resources and data are always available to authorized users when needed.

2. Differentiate between IDS and IPS.

- IDS (Intrusion Detection System): Think of this as a security camera or a burglar alarm. It monitors network traffic for suspicious activity and sends an alert if it finds something, but it doesn't stop the attack itself.
- IPS (Intrusion Prevention System): This is like a security guard. It monitors traffic just like an IDS, but it is proactive—it will actually take action to block or drop the malicious traffic the moment it's detected.

3. Why is a Virtual Private Network (VPN) important?

A VPN is crucial because it creates a secure, encrypted "tunnel" for your data to travel through over the public internet. This prevents hackers or "eavesdroppers" from reading your data even if they manage to intercept it. It also masks your IP address, providing better privacy and anonymity.

4. When would you install a Demilitarized Zone (DMZ)?

You would use a DMZ when you have servers that need to be accessed from the public internet, such as Web servers or Email servers. By placing them in a DMZ, you isolate them from your sensitive internal network. If the public-facing server gets compromised, the DMZ acts as a buffer to prevent the attacker from reaching your private data.

5. What is the role of Network Address Translation (NAT)?

The primary role of NAT is to allow multiple devices on a private network to share a single public IP address to access the internet. From a security standpoint, it hides the internal private IP addresses from the outside world, making it much harder for external attackers to target a specific device inside the network.

6. How does a Proxy Server provide security?

A proxy server acts as an intermediary or "middleman" between a user's device and the internet.

- **Operation:** When a user requests a website, the request goes to the proxy first. The proxy then makes the request on behalf of the user, masking the user's IP address.
- **Security:** It can filter out dangerous content, block unauthorized websites, and provide anonymity. Since the internet only sees the proxy's IP, your internal network structure remains hidden.